STANDARD VIII — CHEMISTRY (ANSWER KEY)

FIRST TERM SUMMATIVE ASSESSMENT 2026 | EM

Subject: Chemistry | Class: Standard 8 | Time: 40 Minutes | Total Score: 20

Q. No.	Value Points / Expected Answers (Answer Only)	Score
1	**(d) Rusting of iron is a chemical change.** (d)	1
2	**B. 1-b, 2-c, 3-d, 4-a** B (1-Metals & Non-metals, 2-Triads, 3-Law of Octaves, 4-Atomic mass)	1
3	**(a)** The cup becomes hot / Heat is felt. **(b)** Heat is liberated during the chemical reaction between quicklime and water (Exothermic reaction). $CaO + H_2O \rightarrow Ca(OH)_2 + Heat$	2 (1+1)
4 (A)	**(a)** A white precipitate of Silver Chloride is formed; it turns black/grey when exposed to sunlight. **(b)** Silver chloride decomposes into metallic silver by absorbing light energy.	2 (1+1)
4 (B)	OR 4 (B) **(a)** Carbon dioxide + Water ($CO_2 + H_2O$) **(b)** Photochemical reaction	2 (1+1)
5	**(a)** Chemical energy → Electrical energy **(b)** Electrical energy → Chemical energy	2 (1+1)
6	• **a.** All matter is made up of extremely tiny particles called atoms. • **c.** Atoms of different elements differ in size, properties, and mass.	2
7 (A)	<table><tr><th>Element</th><th>Basis of Naming</th><th>Symbol</th></tr><tr><td>Indium</td><td>(a) Colour (Indigo)</td><td>In</td></tr><tr><td>(b) Rutherfordium</td><td>Rutherford (Scientist)</td><td>Rf</td></tr><tr><td>Europium</td><td>(c) Continent (Europe)</td><td>Eu</td></tr><tr><td>(d) Neptunium</td><td>Planet (Neptune)</td><td>Np</td></tr></table>	2
7 (B)	OR 7 (B) <table><tr><th>Element</th><th>Symbol</th></tr><tr><td>Aluminium</td><td>Al</td></tr><tr><td>Sulphur</td><td>S</td></tr><tr><td>Gold</td><td>Au</td></tr><tr><td>Potassium</td><td>K</td></tr></table>	2
8	**2N :** Represents 2 separate (individual) nitrogen atoms. **N_2 :** Represents 1 molecule of nitrogen containing 2 bonded nitrogen atoms.	2
9	**(a)** Negative terminal (Cathode) **(b)** Silver cyanide + Sodium cyanide solution (or potassium argentocyanide solution)	3 (1+1+1)

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	(c) Yes. It is an electrochemical reaction because electrical energy is absorbed/consumed to bring about the chemical change.	
10 (A)	• **Number of molecules** = 3 • Number of Carbon atoms = $3 \times 2 = 6$ • Number of Hydrogen atoms = $3 \times 6 = 18$ • **Total number of atoms** = $6 + 18 = 24$	**3**
10 (B)	**OR 10 (B)** **a) Reactant :** $CaCO_3$ (Calcium carbonate) **b) Products :** CaO (Calcium oxide), CO_2 (Carbon dioxide) **c) Chemical equation :** $CaCO_3 \rightarrow CaO + CO_2$ (on heating)	**3** (1+1+1)

Prepared by Ravi Sir